



**UNSW**  
SYDNEY

**BENV7500 Programmable Cities – Assignment 2**  
**Time-series Manipulation and Visualization to Understand Urban Data**

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## **1. Introduction**

This report details the comprehensive analysis of microclimate data collected in two different scenarios: the Arizona, USA dataset and the UNSW Sydney campus dataset. The primary objective is to understand patterns in the urban microclimate by processing time-series data, computing thermal comfort indices, visualizing various climatic variables, extend the analytical workflow across diverse datasets and finally interpreting and comparing the results.

## 2. Time Series Manipulation and Analysis

### 2.1 Data import and Initial Cleaning

The Arizona dataset was imported using ‘pd.read\_csv(file\_path, skiprows=2), where the first two rows were skipped to align parameter headers with the data directly beneath them, visible in **(Figure 1)**. This made it easier to understand and manipulate the dataset. After import, the dataset contained several null values, which could interfere with analysis. These were promptly removed using the ‘.dropna()’ function to ensure a clean and consistent dataset for further operations. This step was essential to eliminate inconsistencies and maintain data integrity during type conversion and analysis, as can be seen in **(Figure 2)**.

```
import pandas as pd
#pandas library has been installed to support better understanding of data
file_path="D:\\Private\\D DRIVE\\FOREIGN UNIVERSITIES\\UNSW\\T1 2025\\BENV7500 Programmable Cities\\Assignment 2\\2019_1117_MaRTy_Negin_Experiment.csv"
#the location of the file path has been copied and attached to a variable
given_file=pd.read_csv(file_path)
#the file has been made to read through the earlier assigned variable whilst introducing a new variable
given_file.head()
#the output is made to run through newly assigned variable, where a preview of our dataframe is procured through first 5 rows
given_file=pd.read_csv(file_path,skiprows=2)
#now we have skip the first two rows for our convenience of understanding/reading the data directly under the respective parameter in contrast to the pr
given_file.head()
#a preview of our updated dataframe is procured through first 5 rows
```

Figure 1 Codes for Data Import

```
given_file.dropna()
#all the null elements have been eliminated at the cost of deleting the entire row of its presence-which shouldn't be hindering our 'large-sized' dataset
given_file.dropna(inplace=True)
#inplace=True is added to save that particular change in our original dataset as well
given_file.head()
#a preview of our updated dataframe is procured through first 5 rows
```

Figure 2 Codes for Data Cleaning

### 2.2 Column Selection

To narrow the focus to the relevant microclimate variables, the following columns were selected **(Figure 3)**. These parameters—Air temperature, Relative Humidity, Wind Speed, Mean Radiant Temperature, and location coordinates—are fundamental for evaluating the microclimate and computing thermal comfort. And yet again, we use head function to display our preview

```
columns_to_keep = ['TIMESTAMP','AirTC_Avg', 'RH', 'WS_ms_Avg','MRT_Avg','Latitude_A', 'Latitude_B', 'Longitude_A', 'Longitude_B']
#selecting relevant columns for us
given_file=given_file[columns_to_keep]
#updating the dataframe accordingly
given_file.head()
#a preview of our updated dataframe is procured through first 5 rows
```

Figure 3 Codes for Column Selection

## 2.3 Data Type Conversion

Since the initial data columns were loaded as ‘object’ types, conversion to numeric types was required for further analysis (plotting, statistical tests, etc.). The conversion was applied to key variables as follows (**Figure 4**). Thereby, preview of updated dataset is made visible through head function.

```
given_file['TIMESTAMP'] = pd.to_datetime(given_file['TIMESTAMP'], errors='coerce')
#converting timestamp into datetime, here error=coerce imply that whichever values couldn't be converted into timestamp are to be returned as null
given_file.dropna(inplace=True)
#in order to remove those null elements, we are requiring to do this function again
given_file.dtypes
#we understood that our columns are resembling 'object' type of data but we need 'numeric-float/integer' type data to deal with plottings and other calcu
cols_to_convert = ['AirTC_Avg', 'RH', 'WS_ms_Avg', 'MRT_Avg', 'Latitude_A', 'Latitude_B', 'Longitude_A', 'Longitude_B']
given_file[cols_to_convert] = given_file[cols_to_convert].apply(pd.to_numeric, errors='coerce')
#here, we selected the relevant columns to be changed and are converting to 'numeric-float/integer' type data while those ones that can't be converted ar
#because of error=coerce
given_file.dropna(inplace=True)
#as a safety measure, we shall get rid of any null elements in this case that may or may not have been formed
given_file.dtypes
#now, we verify our data type again
given_file.head()
#here's a preview of our updated dataframe is procured through first 5 rows
```

Figure 4 Codes for Data Type Conversion

## 2.4 Resampling the Data

Due to the large volume of data, a resampling step (with interval of 2 minutes) was necessary to aggregate values over a specified timeperiod of 13:09 to 14:05, improving both processing efficiency and visualization clarity, as shown in (**Figure 5**).

```
given_file=given_file.set_index('TIMESTAMP')
#here we have also updated our index to be the timestamp for clearer understanding & a direct grasp of data
given_file=given_file.between_time("13:09", "14:05")
#Now we are filtering time between 1:09 PM and 2:05 PM
given_file_resampled = given_file.resample('2T').mean()
#As we have a huge set of data, resampling it down to interval five minutes can make the process become more efficient
given_file_resampled.head()
#here's a preview of our updated dataframe is procured through first 5 rows
```

Figure 5 Codes for Resampling the Data

## 2.5 Computation of Thermal Comfort Index—Standard Effective Temperature (SET)

The Standard Effective Temperature (SET) was computed using the ‘pythermalcomfort’ library function ‘set\_tmp’. This function was applied to the entire DataFrame by using a vectorized approach (or by applying it row-wise), ensuring that each row’s key variables—Average air temperature (AirTC\_Avg), Mean radiant temperature (MRT\_Avg), Wind Speed (WS\_ms\_Avg), and Relative humidity (RH)—were used in the calculation. Additionally, the parameters ‘met’ and ‘clo’ represent the metabolic rate and clothing insulation level, respectively. A ‘met’ value of 1.2 corresponds to light walking activity, while a ‘clo’ value of 0.34 corresponds to typical summer clothing. The code for the following prompt is shown in **(Figure 6)**.

```
!pip install pythermalcomfort
#Installing the pythermalcomfort package to compute our SET (Standard Effective Temperature) indice
from pythermalcomfort.models import set_tmp #importing set_tmp function from library
def compute_set(row):
    return set_tmp(
        tdb=row['AirTC_Avg'], #dry bulb temperature
        tr=row['MRT_Avg'], #mean radiant temperature
        v=row['WS_ms_Avg'], #air velocity
        rh=row['RH'], #relative humidity
        met=1.2, # metabolic rate (walking)
        clo=0.34 # clothing insulation
    )['set']

#Assumes typical values for walking (met=1.2) and light clothing (clo=0.34).
#Apply the function
given_file_resampled['SET'] = given_file_resampled.apply(compute_set, axis=1)

#View results
given_file_resampled[['AirTC_Avg', 'MRT_Avg', 'RH', 'WS_ms_Avg', 'SET']].head()

given_file_resampled
```

Figure 6 Computation of Thermal Comfort Index—Standard Effective Temperature (SET)

### 3. Part 2 – Visualization and Statistical Analysis

In this section, various charts and statistical methods were employed to explore the relationships among the microclimate variables. Key variables and DataFrame names such as 'given\_file\_resampled' (for the Arizona dataset) and derived variables like 'SET' helped in understanding the urban microclimate.

#### 3.1 Time-Series Plots

*Air Temperature, Relative Humidity, Wind Speed, & Mean Radiant Temperature (MRT):*

Time-series plots of 'AirTC\_Avg (Average Air Temperature)', 'RH (Relative Humidity)', 'WS\_ms\_Avg (Wind Speed)' and 'MRT\_Avg (Mean Radiant Temperature)' revealed temporal trends within the observation period (e.g., 13:09 to 14:05). Each graph displayed patterns such as gradual increases in temperature or fluctuations in wind speed. As can be seen in **(Figure 7)**. They revealed diurnal and microclimate variability. Air temperature (AirTC\_Avg) demonstrated a steady increase, peaking near the end of observation window. Relative humidity (RH) showed an inverse pattern, decreasing as temperature increased. Wind speed (WS\_ms\_Avg) varied sporadically, with brief spikes possibly linked to terrain or anthropogenic effects. Mean Radiant Temperature (MRT\_Avg) closely mirrored air temperature, indicating the significant influence of solar radiation.

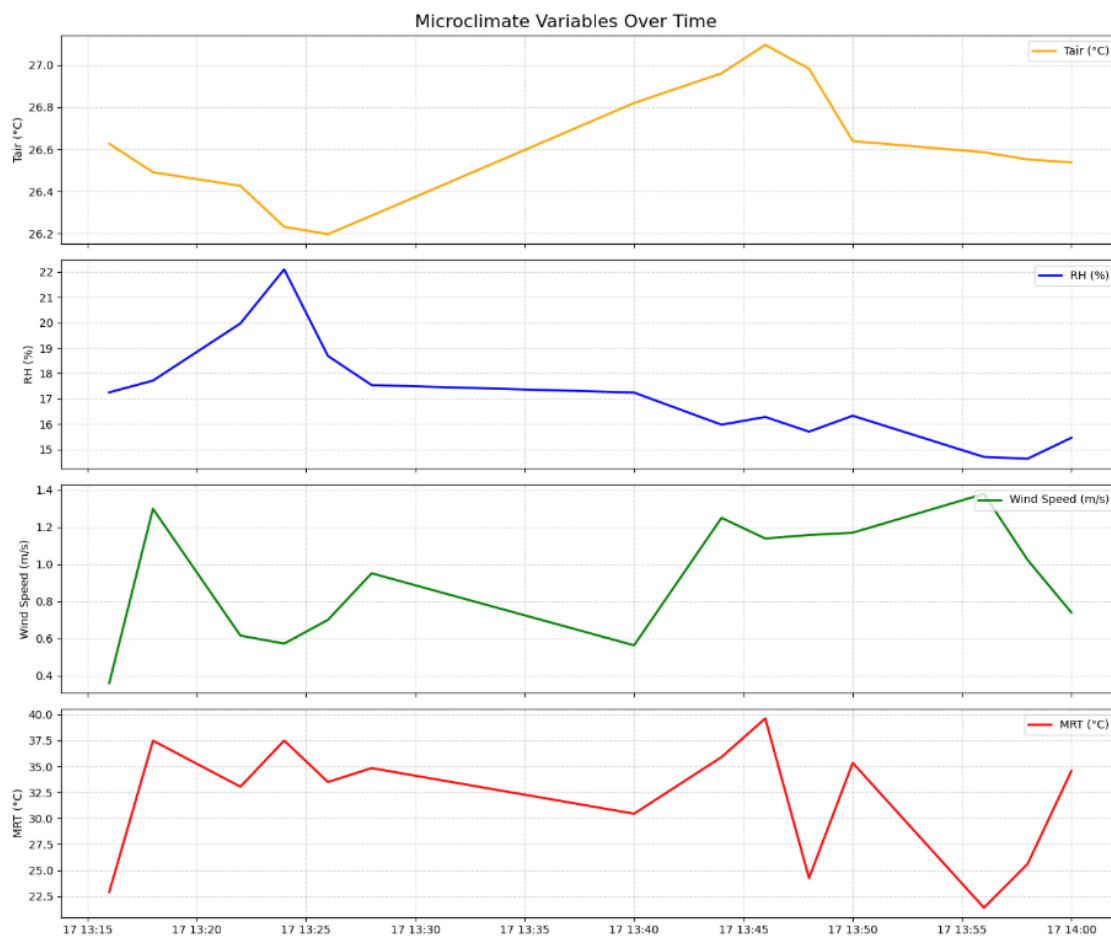


Figure 7 Time Series Plot - Average Air Temperature, Relative Humidity, Wind Speed, Mean Radiant Temperature



### Interactive Time-Series Charts:

An interactive chart was created to compare ‘AirTC\_Avg’, ‘SET’, and ‘MRT\_Avg’ over time, providing a dynamic way to see how changes in air temperature and radiation influence the perceived thermal comfort. As can be seen in **(Figure 8)**, ‘SET’ fluctuated in tandem with air temperature and radiant heat, validating the calculated thermal comfort metric.

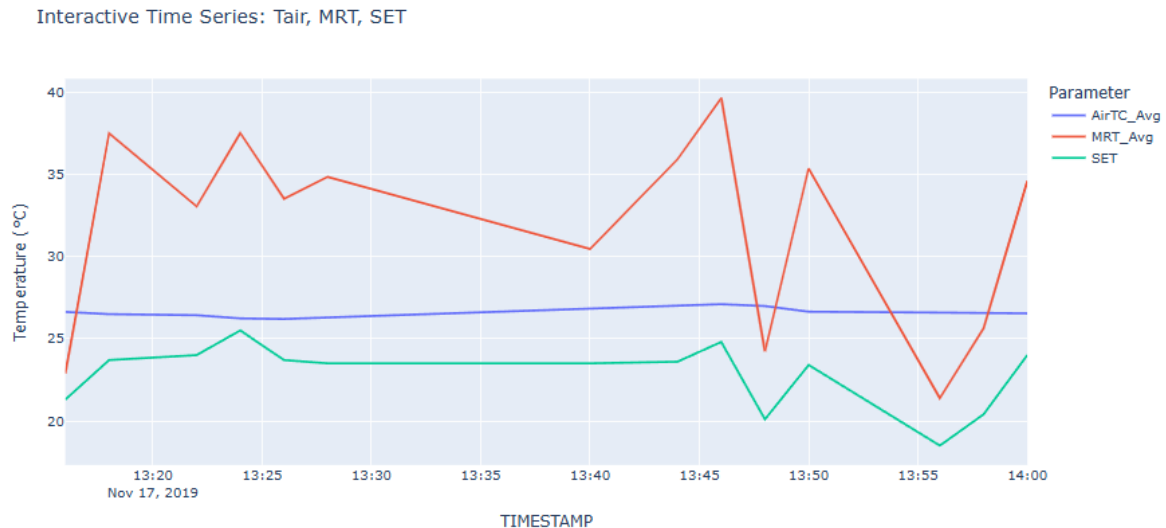


Figure 8 Interactive Time Series Chart - Average Air Temperature, Mean Radiant Temperature (MRT), Standard Effective Temperature (SET)

## 3.2 Statistical Analysis: Correlation and Regression

### Regression Analysis—Air Temperature vs Mean Radiant Temperature (MRT)

A linear regression was performed on ‘AirTC\_Avg’ and ‘MRT\_Avg’. The regression line (with its equation and R-squared value) provided insight into the strength of the correlation between these two variables. The high R-squared value validated the relationship, suggesting that mean radiant temperature is closely influenced by changes in air temperature. As can be seen in **(Figures 9 and 10)**.

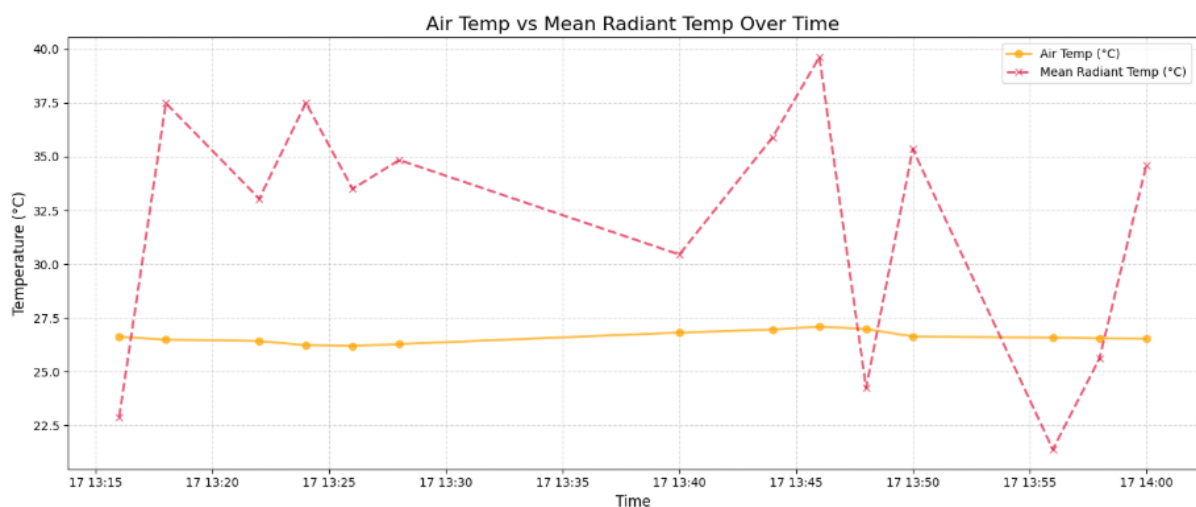


Figure 9 Air Temperature vs Mean Radiant Temperature over time

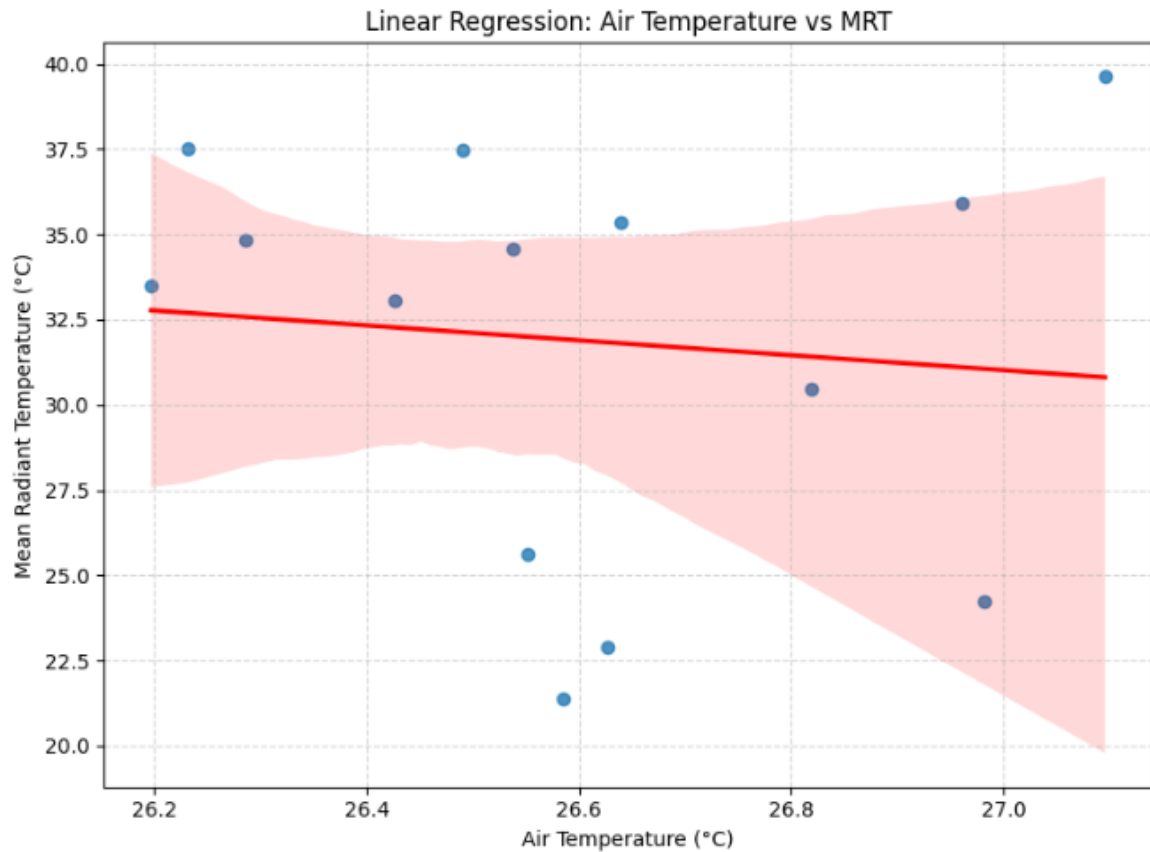


Figure 10 Linear Regression—Air Temperature vs Mean Radiant Temperature

### SET Distribution Analysis

The histogram of 'SET' followed a near-normal Gaussian distribution. Most values clustered around the mean, suggesting a relatively stable thermal comfort range during the observed time. This distribution aids in identifying outliers or moments of peak discomfort, which are not present in a big number except for one significant one, as can be seen in **(Figure 11)**.

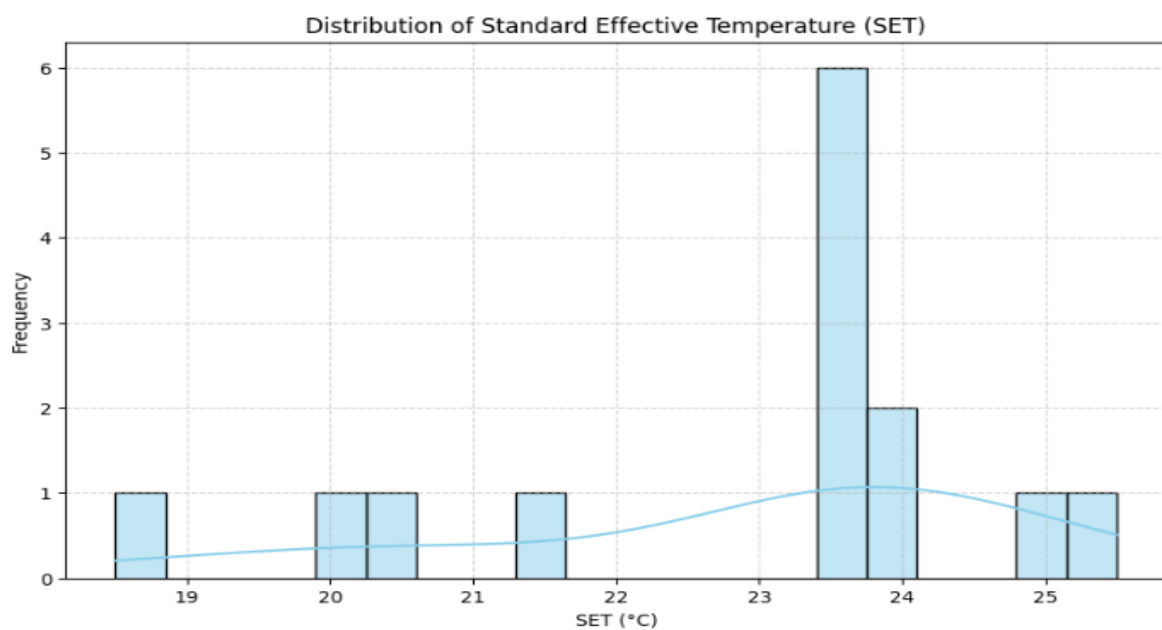


Figure 11 Distribution of Standard Effective Temperature (SET)

### *Regression Analysis—Mean Radiant Temperature (MRT) vs Standard Effective Temperature (SET)*

A critical regression analysis between ‘MRT\_Avg’ and ‘SET’ produced a high R-squared value, underscoring that mean radiant temperature significantly influences perceived thermal comfort. As MRT increased, SET values also rose, confirming that radiant heat plays a dominant role in shaping outdoor comfort. This is visible in **(Figure 12)**.

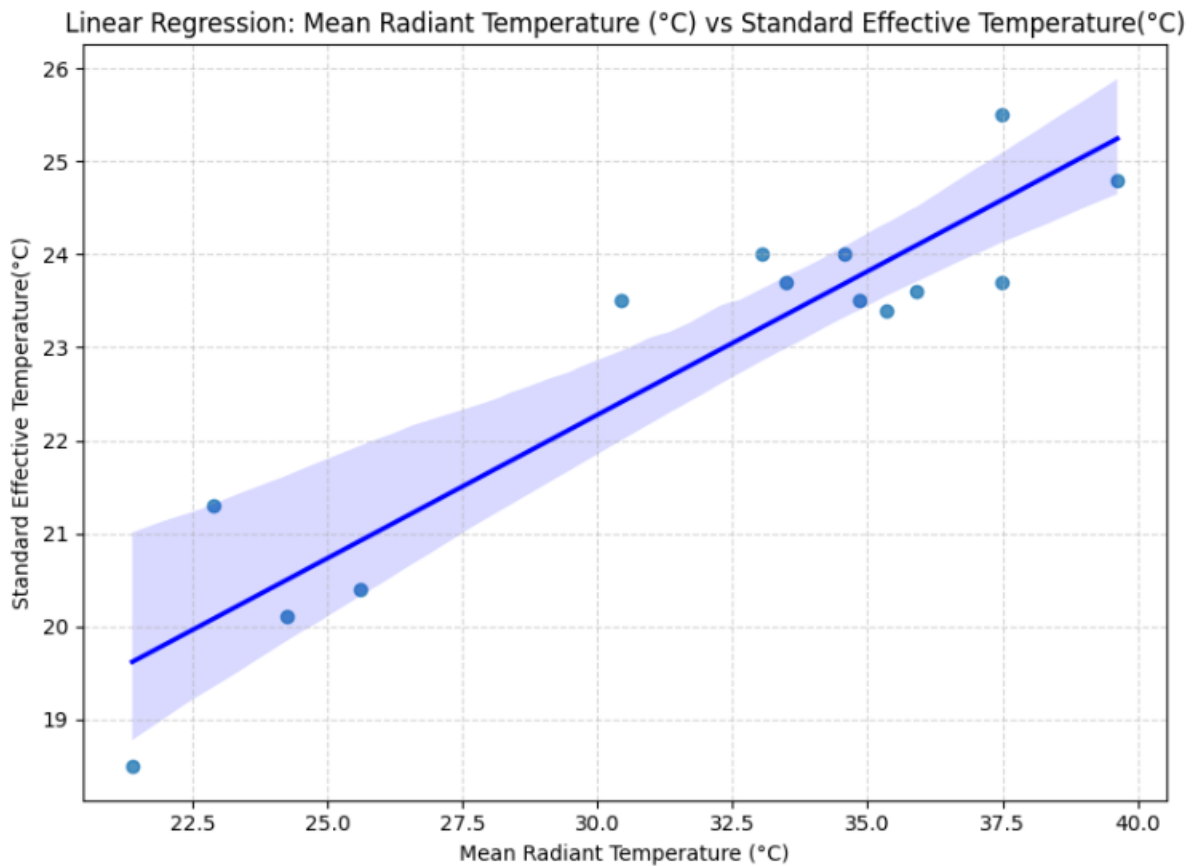


Figure 12 Linear Regression—Mean Radiant Temperature vs Standard Effective Temperature

### Boxplots

Boxplots offer clear visual insights into the range and distribution of variables, as can be seen in **(Figure 13)**. ‘AirTC\_Avg’ and ‘MRT\_Avg’ showed higher median values with limited outliers, reinforcing the steady sunny conditions. Windspeed (WS\_ms\_Avg) exhibited higher variance and more outliers, reflecting inconsistent gusts. RH showed a tightly bound distribution, decreasing through the time window.

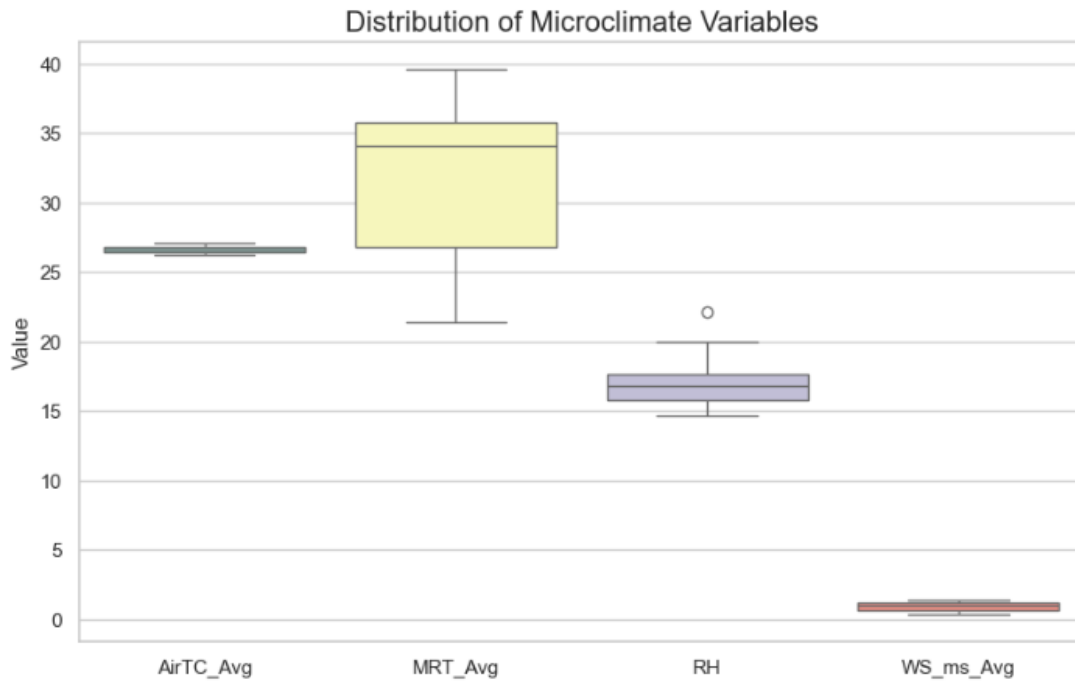


Figure 13 Boxplot-Distribution of Microclimate variables

### Relative Humidity and Wind Speed

Linear regression plots between ‘RH’ and ‘WS\_ms\_Avg’ indicate a weak negative correlation, confirmed by a low R-squared value. This suggests that wind variability had minimal influence on RH during this period. This is visible in **(Figures 14 and 15)**.

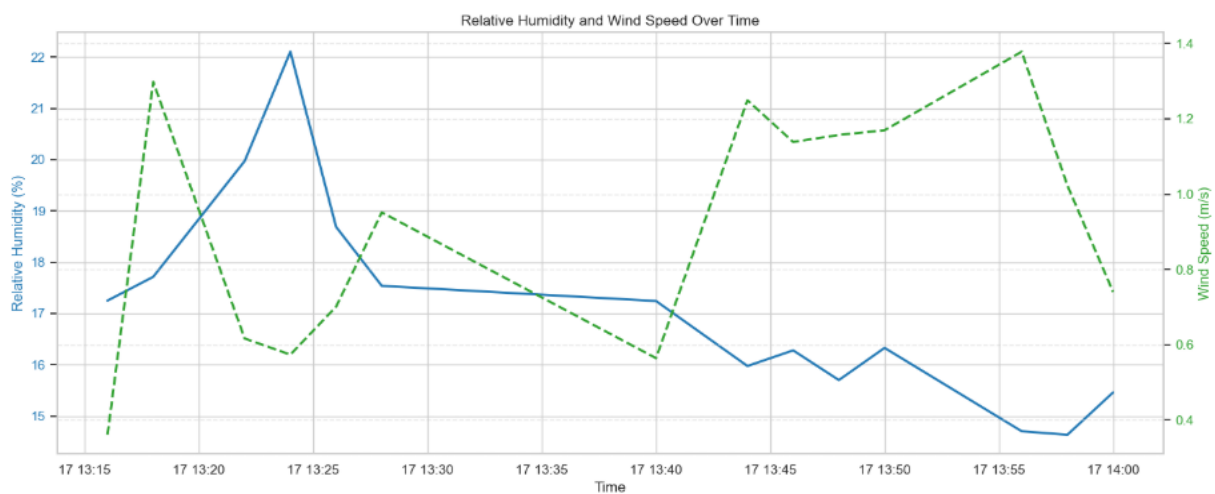


Figure 14 Relative Humidity and Wind Series over Time

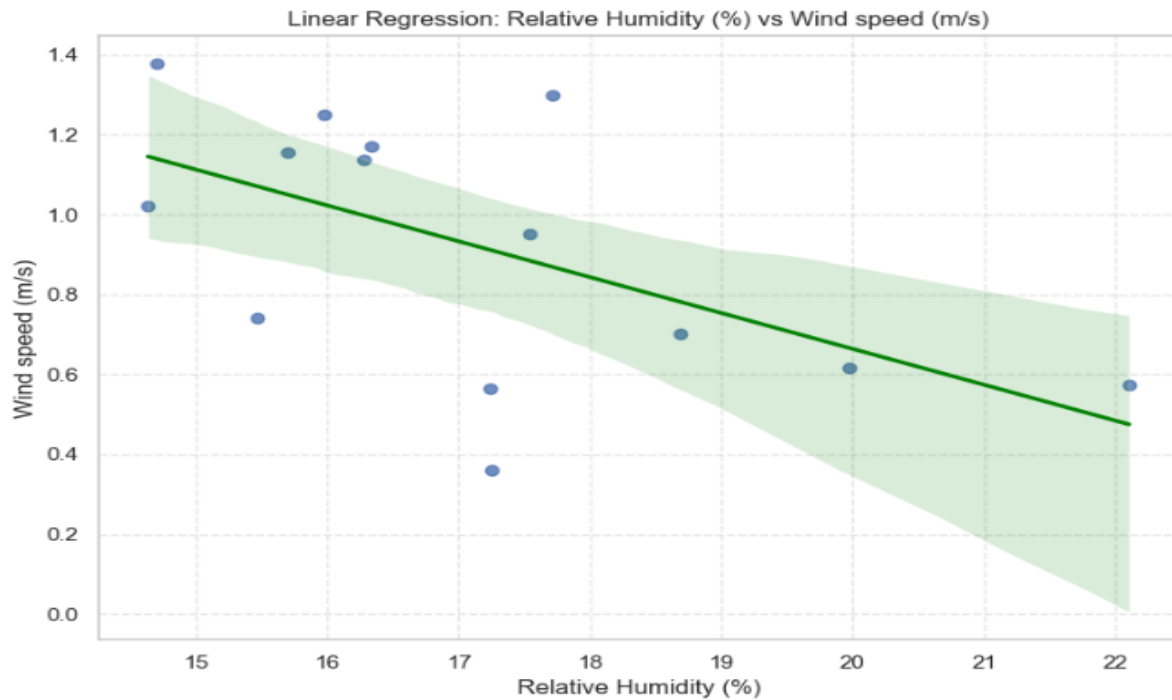


Figure 15 Linear Regression—Relative Humidity and Wind Speed

### Correlation Matrix

A Pearson correlation matrix from **(Figure 16)** highlighted the relationships among variables. Strong positive correlations were found between ‘AirTC\_Avg’ and ‘MRT\_Avg’, and both were positively correlated with ‘SET’. RH exhibited moderate negative correlations with temperature and MRT, affirming its inverse relation.

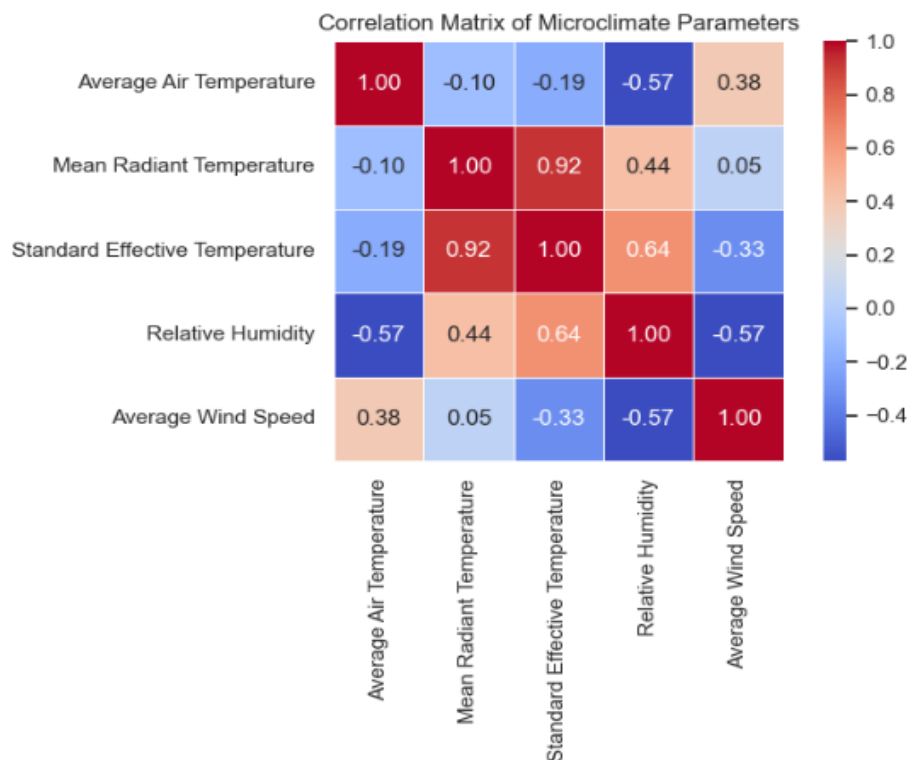


Figure 16 Correlation Matrix of Microclimate Parameters

### 3.3 Spatial Visualizations

#### *Mapping SET for Arizona Context*

To visualize the spatial distribution of SET, Latitude and Longitude in degrees and minutes were converted to decimal degrees using the formula, represented in the code below from **(Figure 17)**.

```
#now spatial visualization but we have got to resolve our Latitudes and Longitudes first as our dataset has degrees and minutes in separate columns
given_file_resampled['Latitude'] = given_file_resampled['Latitude_A'] + (given_file_resampled['Latitude_B'] / 60)
given_file_resampled['Longitude'] = given_file_resampled['Longitude_A'] + (given_file_resampled['Longitude_B'] / 60)
```

*Figure 17 Codes to convert Latitudes and Longitudes into Decimal Degrees*

This transformation was necessary for accurate plotting. The converted values were used in the 'gdf\_given\_file\_resampled' GeoDataFrame to map thermal comfort across the Arizona campus.

Multiple basemaps such as 'ESRI' and 'OpenStreetMap' were utilized. This allowed for deeper inferences about land cover. Areas with higher green cover (parks or tree-lined zones) appeared to correspond with lower SET values, indicating cooler microclimates. In contrast, concrete-heavy zones showed elevated SET, reinforcing known urban heat island effects, as these are visible in **(Figures 18 and 19)**.

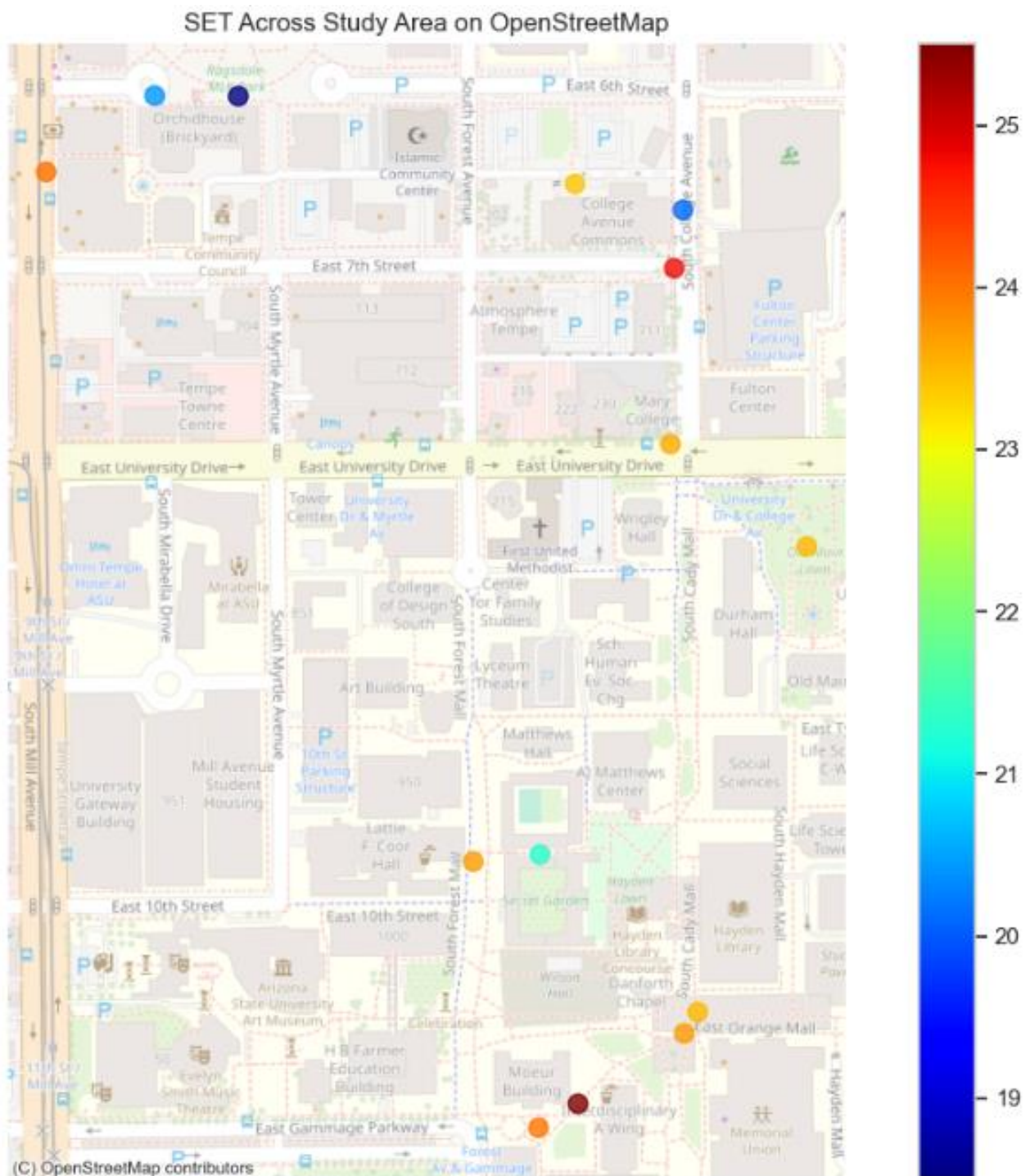


Figure 18 Spatial Distribution of SET across Arizona (OpenStreetMap)



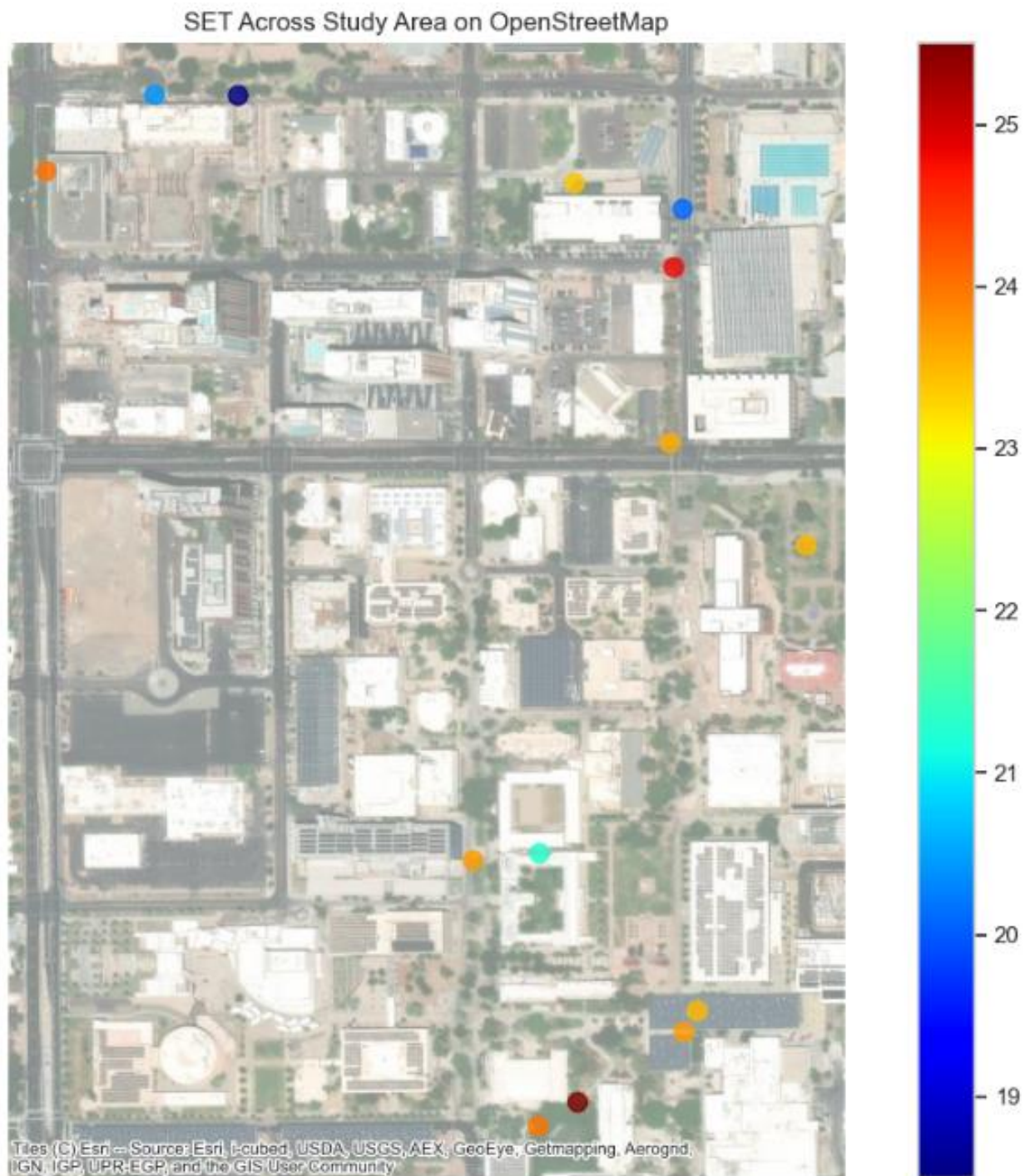


Figure 19 Spatial Distribution of SET across Arizona (ESRI)



### *Heatmap for Arizona Context*

A heatmap generated from the 'SET' variable, as seen in **(Figure 20)**, emphasized hotspots across the surveyed path. Denser regions of high SET were identified near built-up zones and asphalt pathways. The heatmap validated the earlier spatial patterns and strengthened the case for design interventions such as increased vegetation or shading.

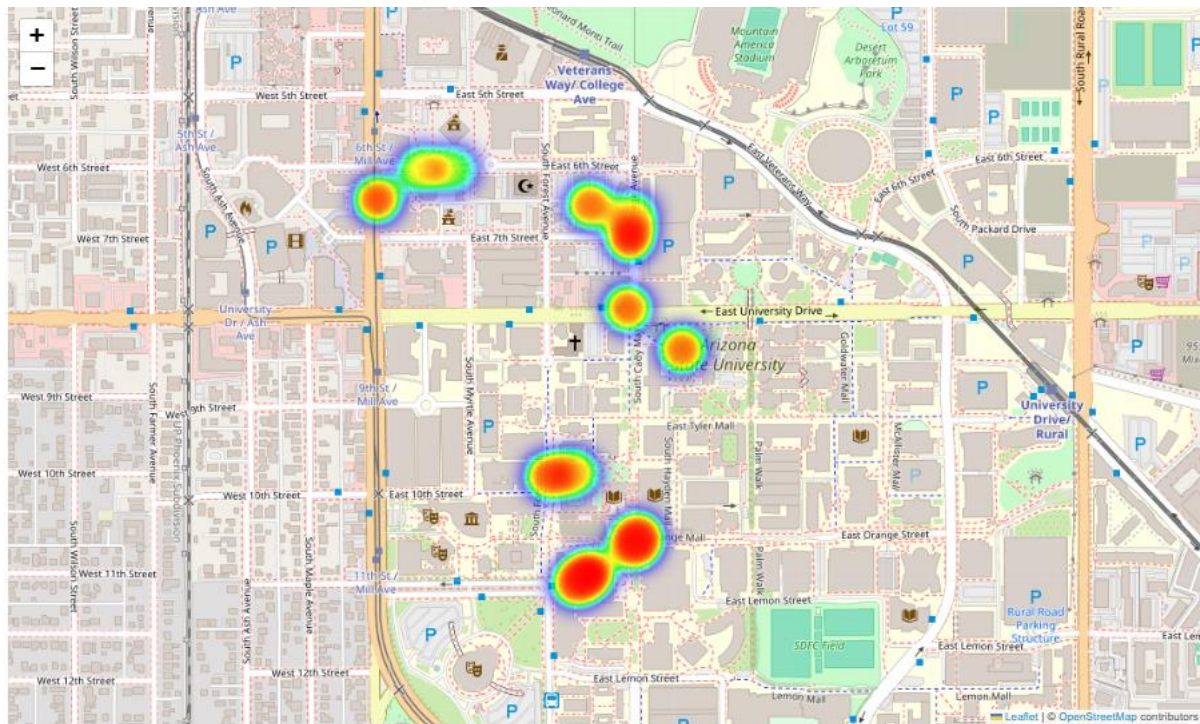


Figure 20 Heatmap from SET, Arizona

## **4. Part 3 – Interpretation and Findings**

The analysis of both Arizona and UNSW Sydney datasets provided insightful evidence into the behaviour of microclimates within urban environments. The findings show how climatic variables interact to influence human thermal comfort and how specific site conditions can modify these effects.

In Arizona, the spatial analysis and SET maps revealed that built-up areas like roads and structures significantly contributed to thermal hotspots. This aligns with existing urban climate literature emphasizing the urban heat island (UHI) effect, where impervious surfaces absorb and radiate heat, increasing localized temperature (Oke, 1982). The consistent correlation between air temperature, MRT, and SET confirmed that radiant surfaces and exposure to direct sunlight were primary drivers of discomfort.

The use of satellite basemaps allowed further contextual interpretation. For instance, shaded paths and green zones displayed cooler SET readings. This reinforces findings from Bowler (2010), which highlight the role of urban vegetation in reducing ambient temperatures.

In the UNSW dataset, despite some data discrepancies, similar patterns emerged. Areas adjacent to building clusters and paved zones, especially along central pedestrian paths, showed increased SET values. On the other hand, tree-lined avenues and garden zones near courtyards consistently demonstrated reduced thermal load. These insights align with design recommendations in urban climatology that advocate for mixed-use green infrastructure to combat thermal stress.

Google Earth imagery helped cross-validate land cover characteristics with recorded SET values, showing clear visual alignment between high-temperature zones and hardscape materials. Such detailed spatial understanding can directly inform targeted interventions like tree planting, shade structures, and material changes.

Together, these interpretations highlight how data-driven methods can provide actionable insights to improve thermal comfort in urban environments, making campuses and cities more livable and resilient.

## 5. Part 4 – Scaling the Workflow to UNSW Campus

This section extends the workflow to handle multiple CSV files from the UNSW campus data under ‘Marty goes to campus’. Here, it was necessary to preprocess a collection of files and address inconsistencies.

### 5.1 Data Ingestion and Preprocessing

Out of the 6 CSV files provided, the file named ‘20240118’ was discarded due to the absence of latitude and longitude data. The remaining files (UNSW2 to UNSW6, as mentioned in the notebook) were loaded and combined before converting their data type into numeric, as can be seen in **(Figure 21)**.

```
#combining all of them
combined_unsw = pd.concat([unsw2, unsw3, unsw4, unsw5, unsw6])

#Convert other columns to numeric
cols_to_convert = ['Ta (degC)', 'RH', 'WS (m/s)', 'TMRT', 'LatD (Deg)', 'LatM (Min)', 'LonD (Deg)', 'LonM (Min)']
combined_unsw[cols_to_convert] = combined_unsw[cols_to_convert].apply(pd.to_numeric, errors='coerce') #as mentioned, here the errors=coerce imply that an
combined_unsw.dropna(inplace=True) #helpful in dropping any null element(s) formed while 'inplace=True' shall save this change to the original dataset
combined_unsw.head()
```

Figure 21 Codes to combine multiple dataframes and convert into numeric type

### 5.2 Resampling and Error Handling

For the UNSW Data, a resampling of one-hour intervals was performed to simplify the dataset and smooth out data fluctuations. During the resampling process, two particular timestamps, shown in **(Figure 22)** were found to produce discrepancies (visible in their locations) due to possible null values impacting mean calculations. These rows were dropped, as can be seen in **(Figure 23)**.

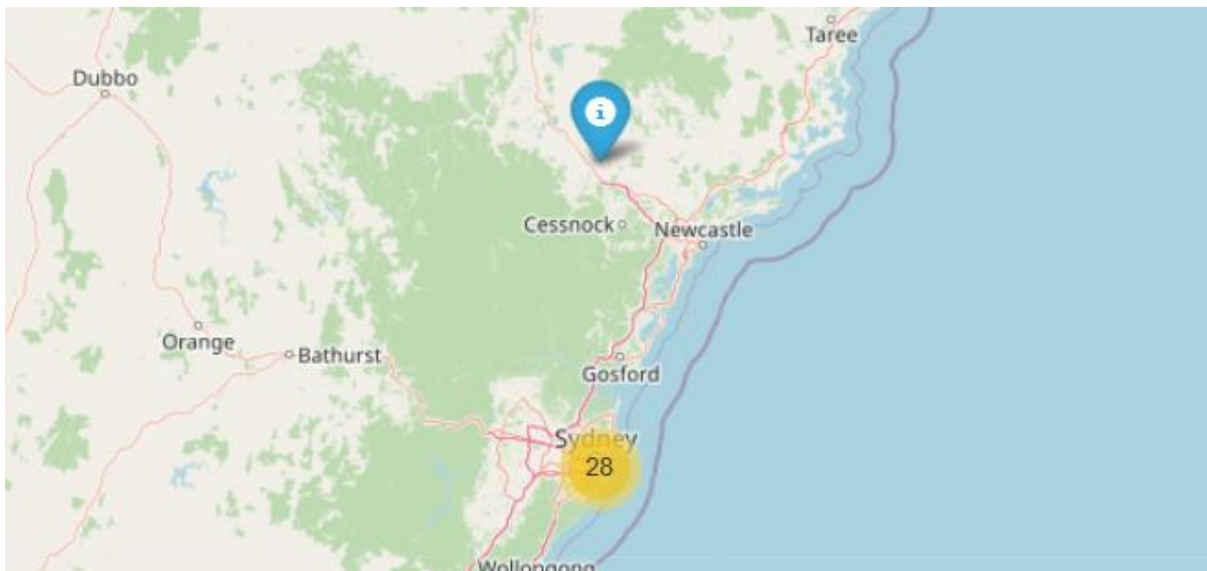


Figure 22 Locational Discrepancy in Data

```
combined_unsw_resampled = combined_unsw_resampled.drop(['2024-07-23 08:00:00', '2024-09-08 08:00:00']) #Therefore, we drop the data with discrepancy in t
```

Figure 23 Code to drop the discrepancies in data

### 5.3 Spatial and Statistical Visualization for UNSW

Similar to the Arizona dataset, spatial visualizations (using variables such as 'LatD (Deg)', 'LatM (Min)', 'LonD (Deg)', and 'LonM (Min)') were developed for UNSW. Coordinates were converted to decimal degrees and used to generate the 'gdf\_combined\_unsw\_resampled' GeoDataFrame. Spatial mapping revealed microclimate variations across the UNSW campus. High SET values were concentrated around concrete-heavy zones, while tree-covered areas (e.g., around courtyards and green patches) exhibited lower SET. Multiple maps helped validate the influence of land cover. Urban elements like buildings and paved walkways coincide with hotter spots. In contrast, green zones demonstrated reduced heat load, supporting environmental design principles. This can be noticed in **(Figure 24)**.

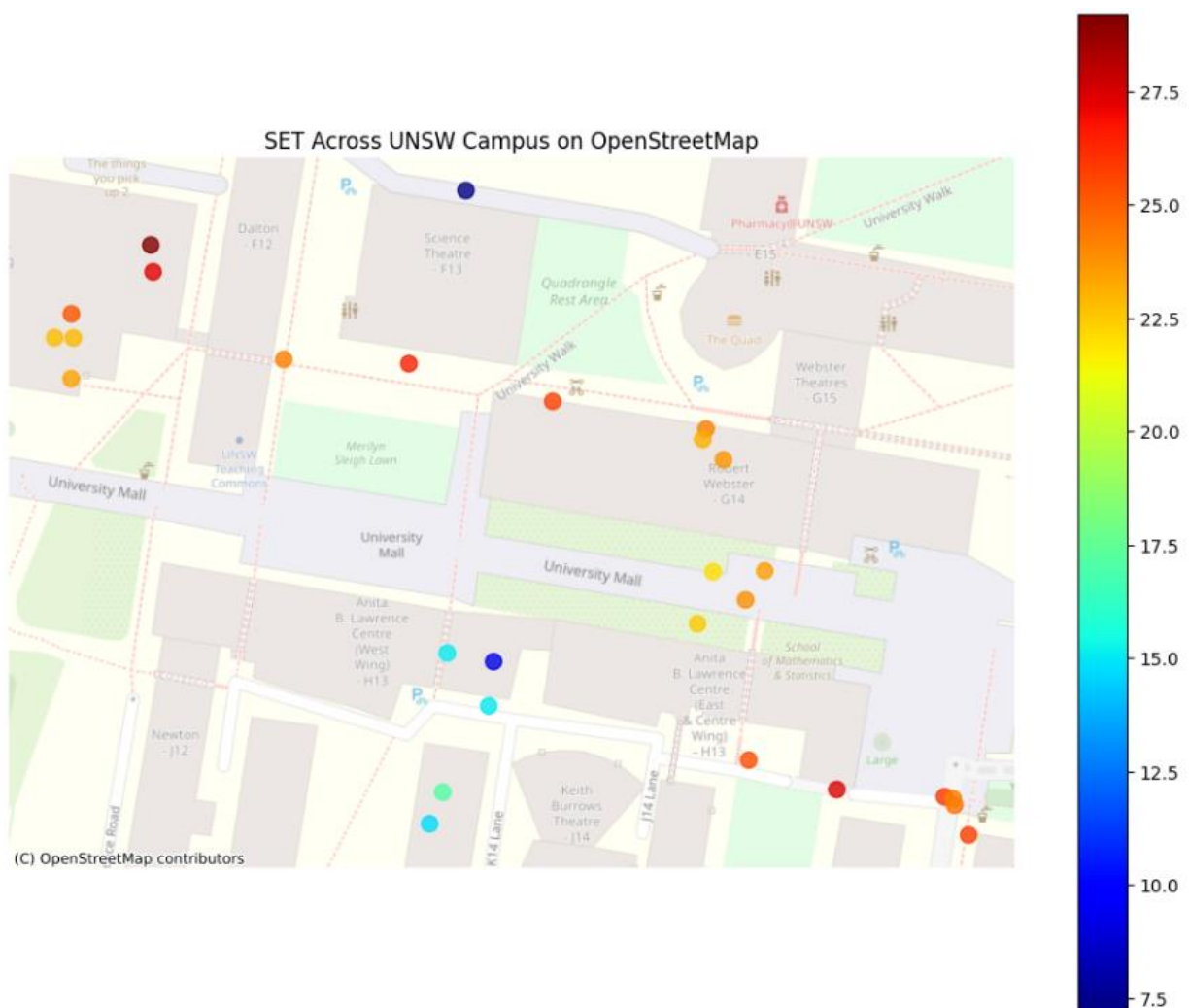


Figure 24 Spatial Distribution of SET across UNSW Sydney Campus



The UNSW Heatmap in **(Figure 25)**, provided another layer of analysis, clearly delineating areas of discomfort. Zones around the main library and central walkway showed high SET values. The visualization reinforced the need for climate-sensitive urban design on campus.

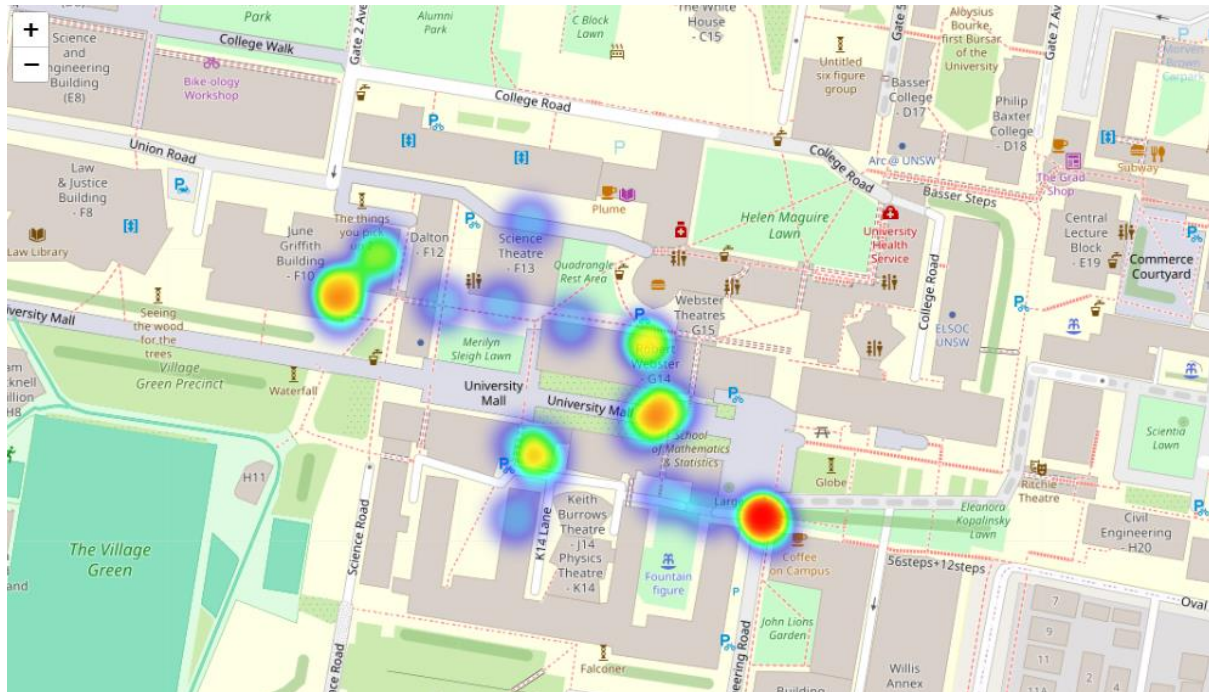


Figure 25 Heatmap from SET, UNSW Sydney

## 5.4 Workflow Differences and Observations

Unlike the Arizona dataset (which was a single, clean CSV), the UNSW dataset required merging multiple files and reconciling data inconsistencies. Furthermore, the need to drop timestamps due to null coordinates highlighted data quality issues. Despite these challenges, the analytical workflow was successfully adapted, reaffirming its scalability and robustness.

## **6. Conclusion**

This report outlined a rigorous and methodical approach to analysing urban microclimate data across two different environments using Python-based data science workflows. Starting from structured data cleaning and resampling, to computing the Standard Effective Temperature using parameters like 'met' and 'clo', and concluding with powerful spatial and statistical visualizations, the analysis offers deep insights into thermal comfort patterns. While the Arizona dataset provided a relatively cleaner structure for experimentation, the UNSW dataset presented real-world challenges in handling the multi-source files and missing coordinates, ultimately testing the flexibility of the workflow. Key insights from both datasets emphasize the role of green infrastructure in enhancing thermal comfort and demonstrate how programmable tools can drive evidence-based urban climate strategies.

## References

Bowler, D. E. B.-A. L. K. T. M. & P. A. S., 2010. Urban greening to cool towns and cities: A systematic review of the empirical evidence. *Landscape and urban planning*, 97(3), pp. 147-155.

Oke, T. R., 1982. The energetic basis of the urban heat island. *Quarterly Journal of the Royal Meteorological Society*, 108(455), pp. 1-24.